Microelectronics Systems – 2023-24

Semester 2

Individual Project Report Template

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Abstract

In this project, an OLED screen was developed to display a specific image in pixels. The entire project was comprised of an L432KC micro-controller and an OLED screen. Many challenges were encountered, such as incorrect libraries. However, the solutions to these problems were flexible. There were three main steps in completing the project, which I will discuss in my introduction. Ultimately, the project achieved its goal of displaying an image on the screen in pixel format.

Introduction

As the MS lecture taught, it is necessary to build something useful, such as a screen to display a certain content. This is a way to convey information through the use of content and images.

In this regard, the following steps are required:

First, find the OLED library to use. The print function is essential. If there are bugs in the libraries but not related to the print function, it is acceptable to ignore them. After obtaining the library, include or cite the library files at the beginning of the code.

Next, the image that is to be displayed should be processed using a modulus operation. After that, the image can be converted into an array that will be sent to the OLED screen to display a pixel-based image.

Finally, complete the final code and compile it for the microcontroller. Then, the image will be presented on the screen in a pixel-by-pixel format.

The code must control all the functional components mentioned above.

Methodology

The preliminary analysis of the problem

An OLED screen that performs an image seems not so difficult to complete. Actually, there was plenty of questions need to be solve. The MBED cannot understand the image if it is a picture.It was necessary to turn it into the language that the MBED can understand, which was called image function. Before using the image function, it should be declared in the library to let the system know your function that can perform an image.

The tools to complete the project

To complete the OLED scree, an 128x64 OLED screen, an L432KC micro-controller, a breadboard piles of wires were required.

Besides, a software that can do modulus operation was required to do the most important step.

Experimental Procedure:

1. Find a proper library in github that support OLED screen

2.Perform a modulus operation

To ensure that the OLED screen can accurately display an image, several crucial steps must be followed. Since the OLED screen cannot directly process a picture as is, we first need to convert it into a format that the micro-controller can interpret.

The first step is to resize the picture to match the dimensions of the OLED screen using image editing software like Photoshop (PS). This ensures that the image will fit perfectly on the screen without any distortion or cropping.

Once the image is resized, it needs to be converted into a format that the microcontroller can understand. This is where the modulus operation comes in. We use special software that performs the modulus operation on the image, translating it into a series of numerical codes. These codes, typically stored in an array format, represent the different pixels and their colors in the image.

After the image has been converted into an array, we can then incorporate it into the code for the micro-controller. The micro-controller reads these numerical codes and uses them to control the OLED screen, turning on and off individual pixels to reproduce the original image. This allows us to display the picture on the OLED screen in a pixel-by-pixel format, ensuring a clear and accurate representation of the original image.

```
const uint8_t Cutedoggyboy[] = {
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0xE0,0x03,0x03,0x3B,0xA2,0x24,0xC8,
    0xD0,0xB0,0xE0,0xC0,0x40,0x40,0x60,0x28,0x30,0x30,0x20,0x30,0x30,0x10,0x10,0x10,
    0x1E,0x1C,0x58,0x70,0xF0,0x98,0x04,0x02,0x01,0xE0,0x7C,0x07,0xFC,0x20,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x09,0xF0,0xF0,0x18,0x0F,0x27,0x90,0x02,0x41,0x60,0x20,
    0xB0,0x11,0x00,0x60,0x00,0x00,0xC0,0x78,0x00,0x00,0x40,0x40,0x60,0x00,0x00,0x00,
    0x04,0x09,0x03,0x0E,0x2E,0x01,0x00,0x60,0x3F,0x01,0x02,0x26,0x4C,0xBC,0xCC,0x08,
    0x00,0x40,0xE0,0xC0,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x80,0x80,0x40,0x70,0xF8,
    0x64,0x12,0x09,0x04,0xC2,0xE3,0x99,0x8C,0xFE,0x03,0x00,0x06,0x80,0xC0,0x60,0x78,
    0x0E,0x03,0x01,0x07,0x1C,0x30,0x60,0x80,0x00,0x00,0x00,0x06,0x0C,0x10,0x20,0x02,
    0x3C,0x80,0x80,0x32,0x02,0x1C,0xF0,0x80,0x00,0x60,0xED,0x07,0x1C,0xF8,0x30,0x20,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0xC0,0x7F,0x9F,0xE0,0xE0,0xFE,
    0xFB,0x78,0xEC,0x70,0x06,0x30,0x3D,0x1F,0x03,0x00,0x00,0x40,0xE3,0xAF,0x93,0xF2,
    0x76,0x3D,0x08,0x11,0xF7,0xFC,0x01,0x02,0x88,0xBB,0x36,0x60,0x88,0x83,0x08,0x10,
    0x10,0xC0,0x00,0x9F,0x00,0x00,0x8F,0xC0,0x10,0x3F,0x38,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x01,0x00,0xFF,0x07,0x03,0x01,0x00,0x00,
    0x00,0x00,0x20,0x20,0x00,0x00,0x00,0x00,0x02,0x03,0x05,0x02,0x00,0x20,0x30,0x1E,
    0x03,0x38,0x14,0x1A,0x2F,0xF0,0xB0,0x27,0x45,0x38,0x30,0x80,0x81,0x60,0xF8,0xC7,
    0x30,0x18,0x0F,0x03,0x03,0x03,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x03,0x06,0x08,0x18,0x30,0x60,0x40,0x40,0x40,
    0x40,0x00,0x00,0xC0,0xC0,0x40,0x60,0x20,0x20,0x20,0x00,0x00,0x00,0x00,0x00,0x06,
    0x01,0x06,0x72,0xF2,0x7E,0x1F,0x1F,0x11,0x03,0x01,0x00,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x80,0x60,0x18,0x0C,0x04,0x00,0x00,0x02,0x00,
    0x01,0x02,0x02,0x01,0x01,0x03,0x03,0x03,0x07,0x05,0x0D,0x01,0x02,0x1A,0x12,0xB2,
    0xA2,0x26,0x26,0x46,0x44,0x46,0x4E,0x40,0x84,0x04,0x04,0x84,0x8C,0x08,0x18,0x30,
    0x60,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x0C,0x07,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x01,0x01,0x03,0x02,
    0x02,0x02,0x02,0x02,0x06,0x02,0x02,0x02,0x01,0x01,0x03,0x02,0x04,0x09,0x06,0x08,
};
```

The code of the function that refers to image

3.Supply the library

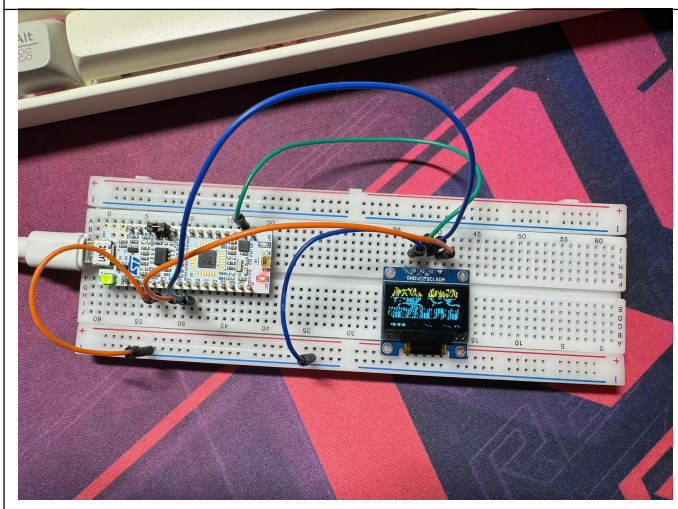
The new function should be added in the library so that the function can be used in main function directly. The function should be declared in the library.

4.Write the main code.

5.Light the screen on

Connect the MBED to the OLED screen using the wire. According to the library, GND pin on MBED and the screen should be connected to ”-” on the bread board. VCC pin connected to 3V3 pin. SCL and SDA were connected to D5 and D4 pin. After building the project, the screen can be light on.

Results and discussion



The figure is the final project that perform a specific image.

To interface the MBED board with an OLED display using wires, based on the library's specifications, you should connect the GND (ground) pin from the MBED and the OLED screen to the '-' or ground rail on the breadboard. The VCC (power supply) pin from the OLED should be connected to the 3V3 pin on the MBED. Furthermore, the SCL (Serial Clock) and SDA (Serial Data) pins from the OLED should be connected to the D5 and D4 pins on the MBED, respectively. Once the circuit is built and the project is programmed, the OLED screen should illuminate.

```
#include "OLED.h"
#include "mbed.h"
#include "OLED_Data.h"

int main()
{
    OLED_Init();
    OLED_ShowImage(1, 1, 120, 60,Cutedoggyboy);
    OLED_Update();
    while(1){}
}
```

The display of the code

As much work done in modulus operation, it was easy to finish the code in 11 rows, including claiming the head, the main function.

Conclusions

The requirement was to build a system that show an image on an OLED screen by using L432KC MBED. By using the proper library and function to show image, it is possible to show image in pixel.

References

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